

NIGAM MEHTA

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Education

Shah & Anchor Kutchhi Engineering College, Mumbai
B.Tech, Electronics Engineering – VLSI Design and Technology

Aug 2023 – 2028

Projects

8-bit Pipelined Processor

GitHub

- Built an 8-bit 3-stage pipelined processor (Fetch/Decode/Execute) in Verilog with an 8-register file, ALU (ADD, ADDI, SUB, SLL, HALT), and hardware hazard detection/stall logic.
- Synthesized and implemented on FPGA (Xilinx Vivado/XSim) at 100MHz with full timing closure, using only 95 LUTs and 104 registers (<2% utilization).

Modular Digital Data Monitoring Unit (SoC IP Core)

GitHub

- Designed an 8-bit RTL monitoring unit with a Moore FSM (Idle, Alarm, Cooldown), sticky alarm, fault capture register, and dual-mode signed/unsigned comparison logic.
- Scaled into an AMBA APB4 slave IP with a 5-register memory-mapped interface; built AutoArchitect in Python to automate Yosys synthesis and netlist SVG generation.

Rail Nova – Railway Reservation Simulator

GitHub

- Built a FastAPI + SQLAlchemy/SQLite reservation system modeling 8,990 stations and 5,277 trains, replicating the CNF/RAC/Waitlist allocation algorithm with auto-promotion on cancellation.
- Added real-time PNR/train tracking via WebSockets, JWT auth, and an in-app e-Wallet.

Credence Core — Digital Banking & Wealth Terminal

GitHub

- Architected a high-performance banking engine in C++20 utilizing `std::unordered_map` and binary-search indexing for $O(1)$ account lookups, integrated via IPC process pipes with a Python Flask REST API.
- Engineered an 8-tab wealth & banking terminal supporting automated Mutual Fund SIPs, 24K digital gold trading, atomic fund transfers, and dynamic certified PDF statement generation with ReportLab.

Professional Experience

Microchip Embedded Systems Developer Intern, AICTE–EduSkills (with Microchip)

Jun – Aug 2026

Completed an 8-week PIC16 microcontroller track: MPLAB X, embedded C (callbacks, modular design), BLE app development, Azure IoT sensor nodes with motor control, and real-time data visualization.

VLSI Design & Semiconductor Engineering Intern, EduSkills Foundation

Apr – Jun 2026

Studied digital IC architectures, RTL design methodologies, ASIC verification flows, and EDA tooling.

Data Analyst Intern, Elevate Labs

Feb – Mar 2026

Cleaned/analyzed 10,000+ records (Python, Pandas, Power BI), built 5 dashboards, and automated reporting – cutting manual effort by 30%.

Android Developer Intern, EduSkills Foundation

Oct – Dec 2025

Developed 3 Android app modules (Java/Kotlin) covering UI screens and core app logic.

AI & Machine Learning Intern, EduSkills Foundation / Edunet Foundation

Jun – Sep 2025

Trained and evaluated ML classification models (scikit-learn) on labeled datasets, reaching 88% accuracy across preprocessing, training, and evaluation stages.

Technical Skills

VLSI/Hardware: Verilog, RTL Design, FSM, ASIC Verification, AMBA APB4, FPGA Synthesis (Xilinx Vivado), Yosys, Digital Logic Design

Embedded: PIC16 MCUs, MPLAB X, Embedded C, BLE/IoT, Azure IoT

Software/AI: Python, C++, SQL, Flask, FastAPI, SQLAlchemy, Scikit-learn, SHAP, SMOTE, Pandas, Power BI, Git/GitHub, Android (Java/Kotlin)

Certifications

Junior Cybersecurity Analyst Career Path (Cisco) | Generative AI in Action (IBM SkillsBuild) | Cybersecurity Fundamentals (IBM SkillsBuild) | Embedded System Design Overview (Maven Silicon) | Full list on portfolio (19 total)